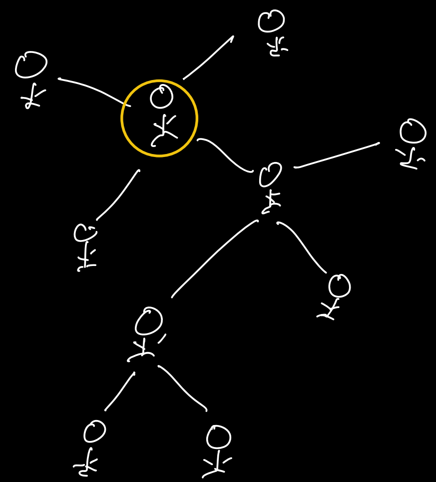
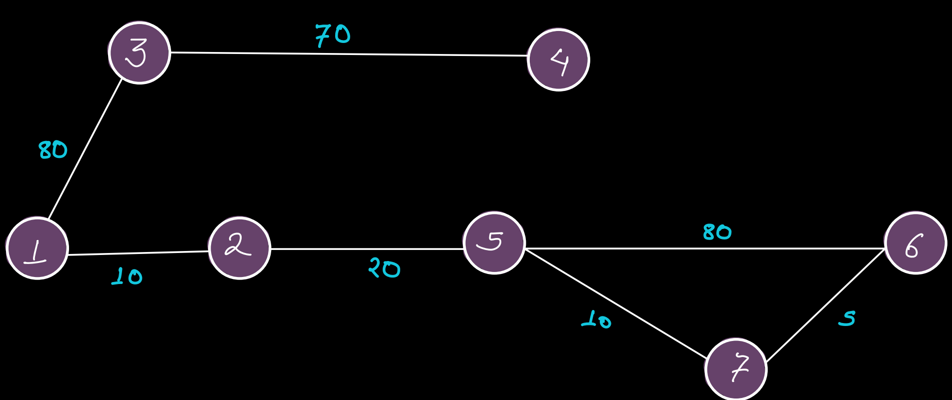


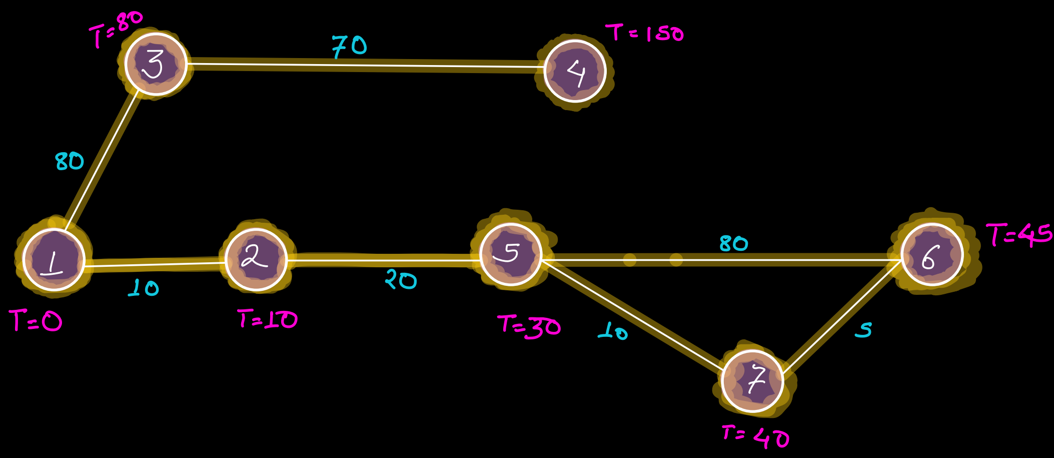
Q Petrol Bunks

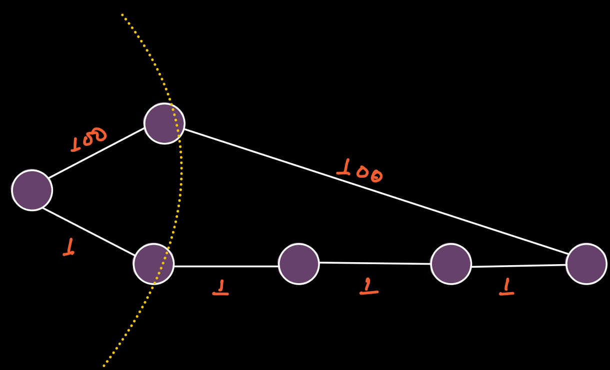
Given the network of petrol bunks with distance b/w them. One of the bunks catches fire.

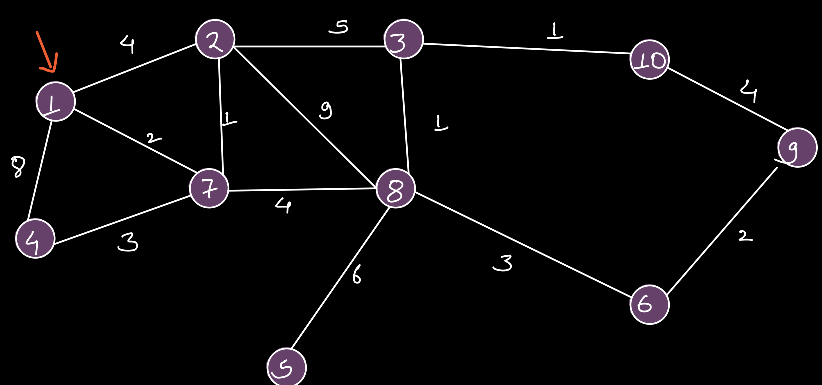
Petrol burns at 1 Km/min.

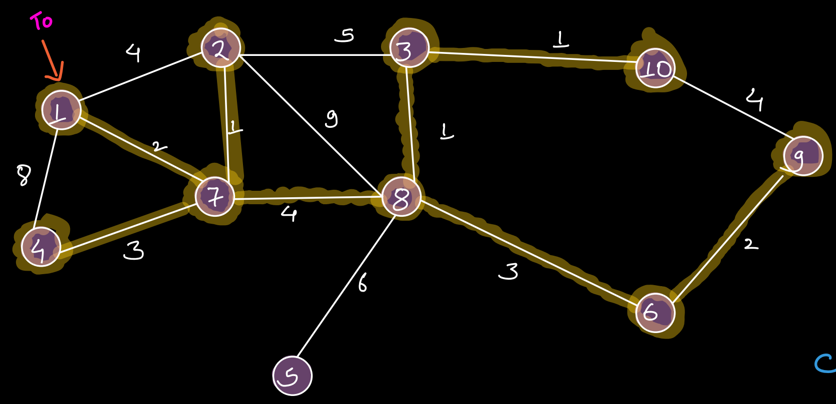
For every bunk, calculate min amount of time after which it will burn down.




T=80
70
T=150
80
T=0
10
T=10
20
T=30
10
80
T=45
5
T=40
1
2
3
4
5
6
7


100
100
1
1
1
1


4
5
1
4
2
9
1
1
8
2
4
3
6
3
2
1
2
3
4
5
6
7
8
9
10



Node	T
1:	0
4 :	~~8~~ 5
2 :	~~4~~ 3
7 :	2
8 :	6
3 :	~~8~~ 7
5 :	12
6 :	9
10 :	8
9 :	~~12~~ 11

```
class Info {
    int node,
    int dist;
}
```

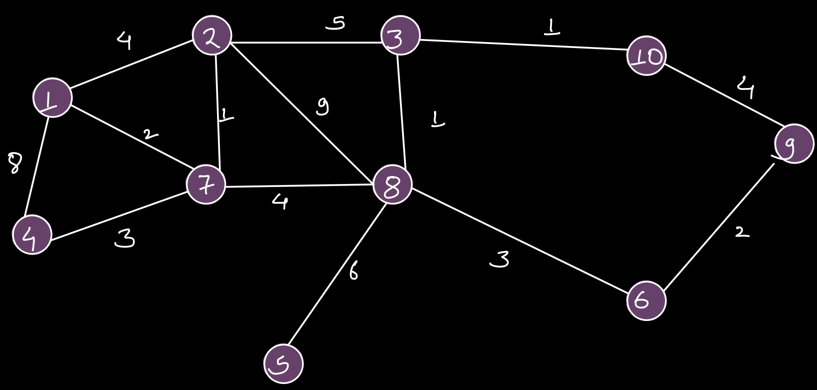
```
dist [N+1] ⟶ ∞
dist [S] = 0;
PQ.add ( {S, 0} )

while ( !PQ.isEmpty() ) {
O(logE) ⟵  temp = PQ.extractMin( )
              u = temp.node;  ⟶ if (u == dest)
                                     ret d
              d = temp.dist;
              for each neighbor v of u :
                  currDistV = dist[u] + getDist(u,v)
                  if dist[V] > currDistV
                        dist[V] = currDistV
O(logE) ⟵              PQ.add ( {V, currDistV} )
```

Size of PQ (max) ⟶ $O(E)$

TC ⟶ $O((V+E)\log E)$

SC ⟶ $O(E + V)$

↑ PQ ↑ dist array



0	1	2	3	4	5	6	7	8	9	10
∞	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~	~~∞~~
	0	~~4~~ 3	~~8~~ 7	~~8~~ 5	12	9	2	6	~~12~~ 11	8

{1, 0} ✓
{7, 2} ✓
{2, 3} ✓
{2, 4} ×
{4, 5}
{8, 6}
{3, 7}
{3, 8} ×
{4, 8} ×
{10, 8}
{6, 9}
{9, 11}
{5, 12}
{9, 12}

Dijkstra's Algo